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SUBJECT OF THE INVENTION

TECHNICAL FIELD The present invention relates to an optical functional integrated device, and in particular, to an optical functional integrated device that performs signal processing with light using an optical waveguide fabricated on the same substrate.

In the broadband era, the introduction of WDM (Wavelength Division Mutiplexing) transmission system enabling communication with multiple optical wavelengths is advanced for efficient utilization of optical fibers. Recently, the use of DWDM apparatus (high density wavelength division multiplexing apparatus) which multiplexes dozens of light wavelengths and enables higher speed transmission is also expanding. Along with this, each WDM transmission system requires a light source corresponding to each optical wavelength, and the required number is dramatically increased with high multiplexing. More recently, commercialization of ROADM (Reconfigurable optical add / drop multiplexers) for adding / dropping an arbitrary wavelength at each node is proceeding. If this system is introduced, in addition to expanding the transmission capacity by wavelength multiplexing, it is possible to switch the optical path by changing the wavelength, and the degree of freedom of the optical network dramatically increases. Furthermore, in the above wavelength multiplex communication system, the number of channels dramatically increases as the transmission capacity increases, and there is also an aspect that control is complicated. In order to individually control dozens of light sources at an advanced level for each channel and realize low price and miniaturization, it is important to integrate optical elements with good controllability in a high density on one chip It becomes essential. PLC (Planar Lightwave Circuit) technology attracts attention as a method to realize a device satisfying this requirement, and various elements have been integrated into one chip so far.

In optical networks, it is desired to miniaturize wavelength tunable lasers, optical multiplexers / demultiplexers, variable optical attenuators, optical switches and the like. Although glass-based PLC devices generally have advantages in terms of integration of optical functions, the size of each device element is still large and it is difficult to integrate multiple functions on one chip. Among others, in order to integrate a variable optical attenuator and an optical switch, it is usual to form a Mach-Zehnder interferometer with an optical waveguide, but at this time, even if it is made of an optical waveguide having a strong optical confinement effect, An element length of about 3 mm is required, and further downsizing is difficult.

Japanese Unexamined Patent Publication No. 03-6504 discloses a light branching element with a light blocking function which can be downsized. The light branching element with a light blocking function is characterized by comprising a silicon substrate, a branched optical waveguide of a quartz type glass material disposed on the substrate, and a light blocking switch disposed in the middle of the branched optical waveguide There. The light blocking switch includes a movable waveguide portion, and the movable waveguide is driven by electrostatic force, thereby passing or blocking the optical signal.

Such an optical branching element with a light blocking function generally has a problem that a high voltage (for example, 500 V) is required. An object of the present invention is to provide an optical functional integrated device that solves the above-described problem of large driving voltage.

An optical functional integrated device according to the present invention comprises: a cantilever beam whose one end is supported by a support side portion; an optical waveguide formed inside the cantilever beam; and an optical waveguide disposed so as to face the other end of the cantilever beam Side optical device and a heater joined to the cantilever beam. The heater is formed of a material having a coefficient of thermal expansion different from the coefficient of thermal expansion of the material from which the cantilever is formed.

According to the present invention, there is provided a method of manufacturing an optical functional integrated device, comprising the steps of: manufacturing a cantilever in which an optical waveguide is formed and one end of which is supported by a support side portion; and arranging so as to face the other end of the cantilever A step of fabricating a heater which is formed of a material having a coefficient of thermal expansion different from a coefficient of thermal expansion of the material on which the cantilever is formed and which is joined to the cantilever beam .

The optical functional integrated device according to the present invention is required when the amount of light traveling between the opposite side optical device and its optical waveguide changes as compared with other optical functional integrated devices that deform the optical waveguide by electrostatic force It is possible to reduce the driving voltage.

BRIEF DESCRIPTION OF THE DRAWINGS The above and other objects, advantages and features of the present invention will become more apparent from the following description in conjunction with the accompanying drawings, with reference to several embodiments.
The accompanying drawings include the following. FIG. 1 is a perspective view showing an embodiment of an optical functional integrated device according to the present invention. FIG. 2 is a perspective view showing a state in the process of manufacturing the optical functional integrated device. FIG. 3 is a perspective view showing another state in the process of manufacturing the

optical functional integrated device. FIG. 4 is a perspective view showing still another state in the process of manufacturing the optical functional integrated device. FIG. 5 is a perspective view showing still another state in the process of manufacturing the optical functional integrated device. FIG. 6 is a perspective view showing another embodiment of the optical function integrated device according to the present invention. FIG. 7 is a perspective view showing still another embodiment of the optical function integrated device according to the present invention. FIG. 8 is a perspective view showing still another embodiment of the optical function integrated device according to the present invention.

Hereinafter, embodiments of the present invention will be described with reference to the drawings.

Embodiments of an optical functional integrated device according to the present invention will be described with reference to the drawings. As shown in FIG. 1, the optical function integrated device 10 includes a silicon substrate 1, a support side portion 2, an opposing side portion 3, a cantilever beam 5, and a thin line heater 6. The silicon substrate 1 is formed of monocrystalline silicon and formed in a plate shape. The support side portion 2 and the opposite side portion 3 are formed of glass and fixed to the surface of the silicon substrate 1. The cantilever 5 is formed of glass and is formed in a rod shape. The cantilever beam 5 is formed such that one end of the cantilever is integrally joined to the support side part 2 and an end face 7 of the other end of the rod faces a part of the opposite side part 3. The cantilever 5 is further warped so that the distance between the silicon substrate 1 and the other end of the cantilever 5 opposite to the one end is larger than the distance between the one end joined to the support side 2 and the silicon substrate 1 .

The thin wire heater 6 is formed of a conductor having a coefficient of thermal expansion larger than the thermal expansion coefficient of the glass. As the conductor, a simple metal such as aluminum Al, chromium Cr, nickel Ni, titanium Ti, tungsten W, gold Au, platinum Pt, a combination of such metals, a conductive thin film exemplified by TiN and TiW Are exemplified. The thin wire heater 6 is bonded to substantially the entire surface 11 of the cantilever 5 opposite to the surface facing the silicon substrate 1. That is, the thin wire heater 6 is joined to the surface 11 from one end joined to the supporting side portion 2 of the cantilever 5 to the other end on the opposite side. The thin wire heater 6 generates heat when a current is applied and thermally expands. When the thin wire heater 6 expands, the cantilever 5 is deformed by the so-called bimetal effect so that the end face 7 approaches the silicon substrate 1. The silicon substrate 1 further has a depression 8 formed in a region where the cantilever 5 is disposed so that the cantilever 5 does not contact the cantilever 5 even if it is deformed.

In the support side part 2, an optical waveguide 14 is formed. That is, the support side portion 2 is formed of an optical waveguide clad and an optical waveguide core. The optical waveguide clad is formed of a material having a refractive index smaller than the refractive index of the optical

waveguide core. The optical waveguide cladding forms the overall outer shape of the supporting side 2. The optical waveguide core is formed inside the optical waveguide cladding, and forms the optical waveguide 14.

The opposite side portion 3 has an optical waveguide 16 formed therein. That is, the opposite side portion 3 is formed of an optical waveguide clad and an optical waveguide core. The optical waveguide clad is formed of a material having a refractive index smaller than the refractive index of the optical waveguide core. The optical waveguide cladding forms the overall contour of the opposite side part 3. The optical waveguide core is formed inside the optical waveguide clad and forms the optical waveguide 16.

The cantilever 5 has an optical waveguide 15 formed therein. That is, the cantilever 5 is formed of an optical waveguide clad and an optical waveguide core. The optical waveguide clad is formed of a material having a refractive index smaller than the refractive index of the optical waveguide core. The optical waveguide cladding forms the overall outer shape of the cantilever beam 5. The optical waveguide core is formed inside the optical waveguide clad and forms the optical waveguide 15.

One end of the optical waveguide 14 is connected to another optical device (not shown) mounted on the optical functional integrated device 10, and the other end is connected to the optical waveguide 15. One end of the optical waveguide 15 is connected to the optical waveguide 14, and the other end thereof is exposed from the end face 7 of the cantilever 5. One end of the optical waveguide 16 is connected to another optical device (not shown) mounted on the optical functional integrated device 10 and the other end thereof is exposed from a surface of the opposite side portion 3 facing the cantilever 5 ing.

The optical function integrated device 10 further includes a control circuit 19. The control circuit 19 is formed separately from the optical functional integrated device 10, and is electrically connected to the thin wire heater 6 via an electric wire (not shown). The control circuit 19 applies a current to the thin wire heater 6 via the electric wire. The control circuit 19 can also be formed on the silicon substrate 1.

2 to 5 show a method of manufacturing the optical functional integrated device 10. In the manufacturing method, first, as shown in FIG. 2, the glass optical waveguide layer 22 is formed on the surface of the silicon substrate 21. The silicon substrate 21 is formed of monocrystalline silicon and is formed in a disc-shaped wafer. In the glass optical waveguide layer 22, the optical waveguide core 23 is formed inside the optical waveguide clad forming the outer shape of the glass optical waveguide layer 22. The optical waveguide core 23 is linearly formed. In the glass optical waveguide layer 22, the thin wire heater 24 is bonded to the surface opposite to the surface bonded to the silicon substrate 21. The thin wire heater 24 is formed in a rectangular shape and

the optical waveguide core 23 in the glass optical waveguide layer 22 is arranged so that the longer side of the rectangle is parallel to the straight line along which the optical waveguide core 23 extends As shown in FIG.

3, the glass optical waveguide layer 22 is then partly formed such that the glass optical waveguide layer 22 is formed on the support side 2, the opposite side 3 and the cantilever 5 A resist 25 is formed. Subsequently, the glass optical waveguide layer 22 is etched by RIE (Reactive Ion Etching) at a portion not covered with the resist 25. By such etching, the glass optical waveguide layer 22 is formed in the support side part 32, the opposite side part 33, and the cantilever part 35, as shown in FIG. 4. Further, the optical waveguide core 23 is formed in the optical waveguide 36, the optical waveguide 37, and the optical waveguide 38 by such etching. The optical waveguide 36 is formed inside the support side portion 32. The optical waveguide 37 is formed inside the cantilevered beam portion 35. The optical waveguide 38 is formed inside the opposite side portion 33.

In the silicon substrate 21, a support side portion 32, an opposite side portion 33, and a cantilever beam portion 35 are formed, and after resist 25 is removed, isotropic selective etching of silicon is performed. By isotropic selective etching, the silicon substrate 21 is formed on the silicon substrate 39 on which the depressions 28 are formed, as shown in FIG. At this time, the cantilever beam portion 35 is separated from the silicon substrate 39 to have a cantilever structure. Since the thermal expansion coefficient of the glass of the cantilever portion 35 is smaller than the thermal expansion coefficient of the thin wire heater 24, the cantilevered beam portion 35 warps in a direction away from the silicon substrate 39 due to the influence of residual stress at the time of film formation of the thin wire heater 24 .

Next, the current of various current values is applied to the thin wire heater 24, and the amount of light traveling between the optical waveguide 37 and the optical waveguide 38 is measured. Based on the measurement result, when the cantilever portion 35 is deformed such that the amount of light traveling between the optical waveguide 37 and the optical waveguide 38 is maximized, the current value of the current applied to the thin wire heater 24 Is calculated, and it is confirmed that no light is transmitted between the optical waveguide 37 and the optical waveguide 38 when no current is applied to the thin wire heater 24.

The control circuit 19 applies a current to the thin wire heater 6 so that light is transmitted between the optical waveguide 15 and the optical waveguide 16 or light is not transmitted between the optical waveguide 15 and the optical waveguide 16. That is, the control circuit 19 applies the obtained current of the predetermined current value to the thin wire heater 6, or does not apply a current to the thin wire heater 6. When a predetermined current is applied by the control circuit 19, the thin wire heater 6 generates heat corresponding to the current value of the applied current and

expands by an expansion amount corresponding to the amount of heat generation. The cantilever beam 5 is deformed by a displacement amount corresponding to the amount of expansion thereof and is deformed so as to maximize the amount of light traveling between the optical waveguide 15 and the optical waveguide 16. When no current is applied to the thin wire heater 6, the cantilever beam 5 is deformed into an initial shape as shown in FIG. 1, and at that time, light is emitted between the optical waveguide 15 and the optical waveguide 16 It deforms so as not to be transmitted. That is, the optical functional integrated device 10 operates as a so-called optical switch.

The optical functional integrated device 10 can also be operated as a variable optical attenuator. At this time, the current of various current values is applied to the thin wire heater 6, and the amount of light traveling between the optical waveguide 15 and the optical waveguide 16 is measured. Based on the measurement result, the attenuation factor when light is transmitted between the optical waveguide 15 and the optical waveguide 16 and the current value of the current applied to the thin wire heater 6 are associated with each other. Based on the correspondence, the control circuit 19 applies a current having a current value corresponding to a desired attenuation factor to the thin wire heater 6, so that the optical waveguide 15 and the optical waveguide 16 are arranged at the desired attenuation factor Light can be transmitted.

Such an optical functional integrated device 10 can be applied to another optical functional integrated device that deforms a cantilever with an optical waveguide formed by using electrostatic force by designing the resistance value of the thin wire heater 24 to an appropriate value In comparison, the cantilever beam 5 can be deformed by applying a relatively low voltage exemplified by 3 V to 5 V to the thin wire heater 6 when operating as an optical switch. In addition, the optical function integrated device 10 has high heat utilization efficiency because the cantilever 5 has a heat insulating shape. Therefore, the optical functional integrated device 10 also has a feature of low power consumption.

The cantilever 5 is preferably short in terms of miniaturization of the device size of the optical functional integrated device 10. Such an optical functional integrated device 10, for example, sufficiently functions as a variable optical attenuator and an optical switch even if the length of the cantilever 5 is about 200 μm, thereby sufficiently reducing the driving voltage. That is, the cantilever beam 5 can be formed in a region having a length of 1/10 or less of the conventional element in the silicon substrate 1 and a width of about 50 μm including the glass etching region. Therefore, such an optical functional integrated device 10 is suitable for fabrication of small optical devices and high integration. In addition, such optical function integrated device 10 also attenuates the light intensity by physically shifting the optical coupling in terms of characteristics, so that a very high extinction ratio can easily be realized.

In such an optical functional integrated device 10, there is no need for the optical axes of the optical waveguide 15 and the optical waveguide 16 to coincide with each other in an initial state in which no current is applied to the thin wire heater 6, It can be done.

The thin wire heater 6 can also be replaced with another thin wire heater formed on a part of the surface 11 of the cantilever 5. Like the optical functional integrated device 10 in the above-described embodiment, such an optical functional integrated device can also reduce the driving voltage. Compared to such an optical function integrated device, the optical function integrated device 10 further has a structure in which the thin wire heater 6 is uniformly expanded and the deformation stress is dispersed throughout the cantilever 5, so that the cantilever 5 It is possible to prevent destruction due to repeated deformation.

The amount of expansion of the thin wire heater 6 corresponds to the amount of heat generation / current, and the cantilever 5 is easy to realize the intermediate state. Therefore, as compared with other optical functional integrated devices which deform a cantilever beam in which an optical waveguide is formed by using electrostatic force, the optical functional integrated device 10 has a structure in which the optical waveguide 15 and the optical waveguide 16 are maximally optically coupled So that the cantilever 5 can be deformed more reliably. The optical function integrated device 10 can more accurately adjust the attenuation factor of the light transmitted through the optical waveguide 15 and the optical waveguide 16. In other words, the optical functional integrated device 10 is particularly suitable for use as a variable optical attenuator.

An end face of the optical waveguide 15 facing the optical waveguide 16 may be formed on a surface not perpendicular to the straight line along which the optical waveguide 15 extends, that is, on a so-called oblique end face. At this time, the optical functional integrated device 10 has an edge surface antireflection effect that reduces the light transmitted from the optical waveguide 14 to the optical waveguide 15 from being reflected by the end face opposing the optical waveguide 16 of the optical waveguide 15 to the optical waveguide 14 . Furthermore, the end face of the optical waveguide 16 facing the optical waveguide 15 can also be formed on a plane not perpendicular to the straight line along which the optical waveguide 16 runs, so-called oblique end face. At this time, the optical functional integrated device 10 exerts an effect of preventing the reflection of the end face, which reduces the light traveling through the optical waveguide 16 from reflecting on the end face facing the optical waveguide 15 of the optical waveguide 16.

The space in which the cantilever 5 is arranged may be filled with matching oil. Such an optical functional integrated device exerts an effect of preventing edge reflection from decreasing reflection of light transmitted from the optical waveguide 14 to the optical waveguide 15 to the end face of the optical waveguide 15 facing the optical waveguide 16. In addition, since the cantilever beam 5 can suppress the unintended displacement by the viscosity of the matching oil, such an

optical function integrated device can reduce the adverse influence when vibrations are applied from the outside.

The optical functional integrated device 10 may further include a spot size converter on an end face of the optical waveguide 16 facing the optical waveguide 15. The spot size converter converts the spot size of the light so that the field of light spreads at the end face portion of the optical waveguide 16. As such a spot size converter, a tapered type, a spreading type, a segment type or the like is exemplified. According to such a spot size converter, it is possible to reduce the loss of light at the time of maximum optical coupling between the optical waveguide 15 and the optical waveguide 16.

FIG. 6 shows another embodiment of the optical functional integrated device according to the present invention. The optical functional integrated device 40 is formed substantially in the same manner as the optical functional integrated device 10 in the above-described embodiment, namely, the silicon substrate 41, the support side portion 42, the opposite side portion 43, the cantilever 45, the thin line heater 46. The silicon substrate 41 is formed of monocrystalline silicon and is formed in a plate shape. The support side portion 42 and the opposing side portion 43 are formed of glass and fixed to the surface of the silicon substrate 41. The cantilever 45 is formed of glass and is formed in a rod shape. The cantilever beam 45 is formed such that one end of the cantilever is integrally joined to the support side portion 42 and an end face 47 of the other end of the rod faces a part of the opposite side portion 43. The cantilever 45 is further warped so that the distance between one end joined to the support side portion 42 and the silicon substrate 41 is larger than the distance between the other end opposite to the one end and the silicon substrate 41 .

The thin wire heater 46 is formed of a conductor having a coefficient of thermal expansion larger than the thermal expansion coefficient of the glass. The thin wire heater 46 is bonded to substantially the entire surface 51 of the cantilever 45 opposite to the surface facing the silicon substrate 41. That is, the thin wire heater 46 is joined to the surface 51 from one end joined to the support side portion 42 of the cantilever 45 to the other end on the opposite side. The thin wire heater 46 generates heat as a current is applied and thermally expands. When the thin wire heater 46 expands, the cantilever beam 45 is deformed by the so-called bimetal effect so that the end face 47 approaches the silicon substrate 41.

In the support side portion 42, an optical waveguide 54 is formed. In the opposite side portion 43, an optical waveguide 56 is formed. The cantilever 45 has an optical waveguide 55 formed therein. One end of the optical waveguide 54 is connected to another optical device (not shown) mounted on the optical functional integrated device 40, and the other end is connected to the optical waveguide 55. One end of the optical waveguide 55 is connected to the optical waveguide 54 and the other end thereof is exposed from the end face 47 of the cantilever 45. One end of the

optical waveguide 56 is connected to another optical device (not shown) mounted on the optical functional integrated device 40 and the other end thereof is exposed from a surface opposed to the cantilever beam 45 of the opposite side portion 43 ing. At this time, the amount of light traveling between the optical waveguide 55 and the optical waveguide 56 is the same as the amount of light traveling between the optical waveguide 37 and the optical waveguide 38 in the above-described embodiment, and the cantilever 45 changes due to deformation.

In the silicon substrate 41, a recess 48 is further formed in a region where the cantilever 45 is disposed, and a pedestal 49 is formed in the recess 48. The pedestal 49 is formed so that the amount of light transmitted between the optical waveguide 55 and the optical waveguide 56 is maximized when the cantilever 45 contacts the pedestal 49.

In manufacturing such an optical functional integrated device 40, a substrate is formed in which a very thin sacrificial layer is previously formed between the silicon substrate and the glass optical waveguide layer. The sacrificial layer is formed of a material that is selectively etched with respect to the glass optical waveguide and the silicon substrate, and a refractory metal film. As a material thereof, PSG (phosphosilicate glass) is exemplified. As the refractory metal, WSi is exemplified. The glass optical waveguide layer is then etched and formed in the support side 42, the opposite side 43 and the cantilever 45. The silicon substrate is isotropically selective etched of silicon after its glass optical waveguide layer is etched. The silicon substrate is subjected to isotropic selective etching, whereby the indentation 48 and the pedestal 49 are formed. The sacrificial layer is selectively etched after the recess 48 and the pedestal 49 are formed in the silicon substrate. By such selective etching, the cantilever 45 is separated from the pedestal 49 and warped in a direction away from the pedestal 49, and is formed in the optical functional integrated device 40.

The optical functional integrated device 60 further includes a control circuit (not shown) for operating the optical functional integrated device 40 as an optical switch. The control circuit applies a current having a predetermined current value to the thin wire heater 46 or does not apply a current to the thin wire heater 46. The thin wire heater 46 generates heat and expands when a predetermined current is applied by the control circuit. At this time, the cantilever 45 is deformed so as to approach the silicon substrate 41 and contacts the pedestal 49. At this time, the amount of light traveling between the optical waveguide 55 and the optical waveguide 56 is maximized. When no current is applied to the thin wire heater 46, the cantilever beam 45 is deformed into an initial shape as shown in FIG. 6, and at this time, the light passes between the optical waveguide 55 and the optical waveguide 56 It deforms so as not to be transmitted. That is, the predetermined current value is sufficiently large so that the cantilever 45 deforms until it contacts the pedestal 49.

It should be noted that the optical functional integrated device 40 can also be operated as a variable optical attenuator. At this time, the current of various current values is applied to the

thin wire heater 46, and the amount of light traveling between the optical waveguide 55 and the optical waveguide 56 is measured. Based on the measurement result, the attenuation factor when light is transmitted between the optical waveguide 55 and the optical waveguide 56 and the current value of the current applied to the thin wire heater 46 are associated with each other. Based on the correspondence, the control circuit applies a current having a current value corresponding to a desired attenuation factor to the thin wire heater 46, so that the optical waveguide 55 and the optical waveguide 56 are arranged at the desired attenuation factor Light can be transmitted.

Like the optical functional integrated device 10 in the above-described embodiment, such an optical functional integrated device 40 can easily realize miniaturization, high integration, low voltage drive, and extremely high extinction ratio it can. Such an optical functional integrated device 40 can further fix the cantilever beam 45 at a position where optical coupling is maximized. Particularly in the case of an optical switch, realization of high-speed optical switching and stable maximum Coupling can be obtained.

The pedestal 49 can also be formed so that the amount of light traveling between the optical waveguide 55 and the optical waveguide 56 becomes a value other than the maximum value when the cantilever 45 contacts the pedestal 49 . For example, the pedestal 49 can be formed so that the amount of light traveling between the optical waveguide 55 and the optical waveguide 56 is minimized when the cantilever 45 contacts the pedestal 49. Like the optical functional integrated device 10 in the above-described embodiment, such an optical functional integrated device 40 can easily realize miniaturization, high integration, low voltage drive, and extremely high extinction ratio it can.

FIG. 7 shows still another embodiment of the optical functional integrated device according to the present invention. The optical functional integrated device 60 is formed in substantially the same manner as the optical functional integrated device 10 in the above-described embodiment, that is, the silicon substrate 61, the support side portion 62, the opposite side portion 63, the cantilever 65, the SiN film 72 and a thin wire heater 66. The silicon substrate 61 is formed of monocrystalline silicon and formed in a plate shape. The support side portion 62 and the opposing side portion 63 are formed of glass and fixed to the surface of the silicon substrate 61. The cantilever 65 is formed of glass and formed into a rod shape. The cantilever 65 is formed such that one end of the cantilever is integrally joined to the support side portion 62 and an end face 67 at the other end of the rod faces a part of the opposite side portion 63. The SiN film 72 is formed on the entire surface of the supporting side portion 62, the opposite side portion 63, and the cantilever 65 opposite to the surface opposed to the silicon substrate 61. The SiN film 72 prevents the end face 67 of the cantilever 65 from warping away from the silicon substrate 61 by applying stress to the cantilever beam 65.

The thin wire heater 66 is formed of a conductor having a coefficient of thermal expansion larger than the thermal expansion coefficient of the glass. The thin wire heater 66 is bonded to substantially the entire surface 71 of the cantilever 65 opposite to the surface opposed to the silicon substrate 61. That is, the thin wire heater 66 is joined to the surface 71 from one end joined to the support side portion 62 of the cantilever 65 to the other end on the opposite side. The thin wire heater 66 generates heat when a current is applied and thermally expands. When the thin wire heater 66 expands, the cantilever 65 deforms so that the end face 67 approaches the silicon substrate 61 due to the so-called bimetal effect.

The support side portion 62 has an optical waveguide 74 formed therein. In the opposite side portion 63, an optical waveguide 76 is formed. The cantilever 65 has an optical waveguide 75 formed therein. One end of the optical waveguide 74 is connected to another optical device (not shown) mounted on the optical functional integrated device 60, and the other end thereof is connected to the optical waveguide 75. One end of the optical waveguide 75 is connected to the optical waveguide 74, and the other end thereof is exposed from the end face 67 of the cantilever 65. One end of the optical waveguide 76 is connected to another optical device (not shown) mounted on the optical functional integrated device 60 and the other end thereof is exposed from a surface facing the cantilever 65 of the opposite side portion 63 ing. At this time, the amount of light traveling between the optical waveguide 75 and the optical waveguide 76 is the same as the amount of light traveling between the optical waveguide 37 and the optical waveguide 38 in the above-described embodiment, and the cantilever 65 changes due to deformation.

The silicon substrate 61 further has a recess 68 formed in a region where the cantilever 65 is disposed. The recess 68 is deeply formed so that the cantilever 65 can deform until the amount of light traveling between the optical waveguide 75 and the optical waveguide 76 is minimized.

In manufacturing the optical functional integrated device 60, first, a glass optical waveguide layer is formed on the surface of a silicon substrate, and a SiN layer (not shown) is formed on the surface of the glass optical waveguide layer opposite to the surface bonded to the silicon substrate Is formed by plasma CVD (chemical vapor deposition). In the plasma CVD, the input RF power, the introduced gas condition, the frequency mixing, and the like are adjusted so that the SiN film 72 applies predetermined stress to the cantilever 65 when the optical functional integrated device 60 is completed. The predetermined stress is set so that the optical axis coincides with the optical axis direction so that the amount of light transmitted through the optical waveguide 75 and the optical waveguide 76 is maximized in the initial state in which no current is applied to the thin wire heater 66 at the completion of the optical functional integrated device 60 As shown in FIG. 3B, the stress that deforms the cantilever 65. The glass optical waveguide layer is then etched and formed on the support side 62, the opposite side 63 and the cantilever 65. The silicon substrate is isotropically selective etched of silicon after its glass optical waveguide layer is etched. The silicon substrate is

isotropically selectively etched to form a recess 68. By such etching, the cantilever 65 is separated from the silicon substrate 61 and formed in the optical functional integrated device 60.

The optical function integrated device 60 further includes a control circuit (not shown) for operating the optical function integrated device 60 as an optical switch. The control circuit applies a current having a predetermined current value to the thin wire heater 66 or does not apply a current to the thin wire heater 66. The thin wire heater 66 generates heat and expands when a predetermined current is applied by the control circuit thereof. At this time, the cantilever 65 deforms so as to approach the silicon substrate 61. At this time, the amount of light traveling between the optical waveguide 75 and the optical waveguide 76 is minimized. The cantilever 65 deforms to an initial shape as shown in FIG. 7 when no current is applied to the thin wire heater 66. At this time, the light transmitted between the optical waveguide 75 and the optical waveguide 76 In order to maximize the amount.

The optical functional integrated device 60 can also be operated as a variable optical attenuator. At this time, the control circuit applies a current having a current value corresponding to a desired attenuation factor to the thin wire heater 66, so that the light is transmitted between the optical waveguide 75 and the optical waveguide 76 at the desired attenuation factor .

Like the optical functional integrated device 10 in the above-described embodiment, such an optical functional integrated device 60 can easily realize miniaturization, high integration, low voltage drive, and extremely high extinction ratio it can. Such an optical functional integrated device 60 further has a function of maximizing the amount of light transmitted through the optical waveguide 75 and the optical waveguide 76 in the initial state in which no current is applied to the thin wire heater 66, It is combined on (normally on).

Note that the SiN film 72 is formed so that the amount of light traveling between the optical waveguide 75 and the optical waveguide 76 becomes a value other than the maximum value in the initial state in which no current is applied to the thin wire heater 66 Also you can. For example, the SiN film 72 can also be formed so that the amount of light traveling between the optical waveguide 75 and the optical waveguide 76 is minimized in an initial state in which no current is applied to the thin wire heater 66. Like the optical functional integrated device 10 in the above-described embodiment, such an optical functional integrated device 60 can easily realize miniaturization, high integration, low voltage drive, and extremely high extinction ratio it can.

FIG. 8 shows still another embodiment of the optical functional integrated device according to the present invention. The optical functional integrated device 80 is formed substantially in the same manner as the optical functional integrated device 10 in the above-described embodiment, that is, the silicon substrate 81, the support side portion 82, the laser chip 83, the cantilever 85, and

the thin line heater 86 . The silicon substrate 81 is formed of monocrystalline silicon and formed in a plate shape. The support side portion 82 is formed of glass and is fixed to the surface of the silicon substrate 81. The cantilever beam 85 is formed of glass and formed into a rod shape. In the cantilever beam 85, one end of the cantilever is integrally joined to the support side portion 82. The cantilever 5 is further warped so that the distance between the silicon substrate 1 and the other end of the cantilever 5 opposite to the one end is larger than the distance between the one end joined to the support side 2 and the silicon substrate 1 .

The laser chip 83 is fixed to the surface of the silicon substrate 81 so as to face the end face 87 at the end opposite to the end joined to the support side portion 82 of the cantilever 85. The laser chip 83 oscillates the laser in response to the input electric signal.

The thin wire heater 86 is formed of a conductor having a coefficient of thermal expansion larger than the thermal expansion coefficient of the glass. The thin wire heater 86 is bonded to substantially the entire surface 91 of the cantilever 85 opposite to the surface facing the silicon substrate 81. That is, the thin wire heater 86 is joined to the surface 91 from one end joined to the support side portion 82 of the cantilever 85 to the other end on the opposite side. The thin wire heater 86 generates heat as a current is applied and thermally expands. When the thin wire heater 86 expands, the cantilever beam 85 is deformed by the so-called bimetal effect so that the end face 87 approaches the silicon substrate 81.

In the support side portion 82, an optical waveguide 92 is formed. The laser chip 83 has an optical waveguide 94 formed therein. The cantilever 85 has an optical waveguide 93 formed therein. The optical waveguide 92 has one end connected to another optical device (not shown) mounted on the optical functional integrated device 80 and the other end connected to the optical waveguide 93. One end of the optical waveguide 93 is connected to the optical waveguide 92, and the other end thereof is exposed from the end surface 87 of the cantilever 85. One end of the optical waveguide 94 is exposed from the surface facing the cantilever beam 85 of the laser chip 83. The laser chip 83 oscillates laser light toward the end face 87 of the cantilever 85 via the optical waveguide 94. At this time, the amount of light traveling between the optical waveguide 93 and the optical waveguide 94 is the same as the amount of light traveling between the optical waveguide 37 and the optical waveguide 38 in the above-described embodiment, and the cantilever 45 changes due to deformation.

The silicon substrate 81 further has a recess 88 formed in a region where the cantilever 85 is disposed. The recess 88 is formed so that the cantilever 85 can be deformed without coming into contact with the silicon substrate 81.

The optical functional integrated device 80 further includes a control circuit (not shown). The control circuit outputs an electric signal to the laser chip 83, oscillates the laser from the laser chip 83, or stops the oscillation of the laser on the laser chip 83. The control circuit further applies a current of a predetermined current value to the thin wire heater 86 or does not apply a current to the thin wire heater 86. The thin wire heater 86 generates heat and expands when a predetermined current is applied by the control circuit thereof. At this time, the cantilever 85 deforms so as to approach the silicon substrate 81. At this time, the amount of laser transmitted from the laser chip 83 to the optical waveguide 93 is maximized. The cantilever 85 deforms to an initial shape as shown in FIG. 8 so that the laser oscillated from the laser chip 83 is not transmitted to the optical waveguide 93 when no current is applied to the thin wire heater 86 .

The optical functional integrated device 80 can also be operated as a variable optical attenuator. At this time, the control circuit applies a current having a current value corresponding to a desired attenuation factor to the thin wire heater 86, so that the laser oscillated from the laser chip 83 is transmitted to the optical waveguide 93 at the desired attenuation factor .

Like the optical functional integrated device 10 in the above-described embodiment, such an optical functional integrated device 80 can easily realize miniaturization, high integration, low voltage drive, and extremely high extinction ratio it can. It should be noted that the laser chip 83 may be replaced with another light emitting element. As the light emitting element, a light emitting diode is exemplified. Further, the laser chip 83 may be replaced with another optical device. Examples of the optical device include an optical fiber and a light receiving element. Like the optical functional integrated device 10 in the above-described embodiment, such an optical functional integrated device can easily realize miniaturization, high integration, low voltage drive, and extremely high extinction ratio .

Note that the cantilever beam 85 is formed so that the amount of light traveling between the optical waveguide 93 and the optical waveguide 94 becomes a value other than the minimum value in the initial state in which no current is applied to the thin wire heater 86 You can also do. For example, the cantilever beam 85 is formed so that the amount of light traveling between the optical waveguide 93 and the optical waveguide 94 is maximized in the initial state in which the SiN film is formed and no current is applied to the thin wire heater 66 , Can also be formed. Like the optical functional integrated device 10 in the above-described embodiment, such an optical functional integrated device 80 can easily realize miniaturization, high integration, low voltage drive, and extremely high extinction ratio it can.

An end face of the optical waveguide 93 facing the optical waveguide 94 may be formed on a surface not perpendicular to the straight line along which the optical waveguide 93 extends, that is, on a so-called oblique end face. At this time, the optical functional integrated device 80 has an

edge surface antireflection effect that reduces the light transmitted from the optical waveguide 92 to the optical waveguide 93 to the end facing the optical waveguide 94 of the optical waveguide 93 to be transmitted to the optical waveguide 92 . Furthermore, the end face of the optical waveguide 94 facing the optical waveguide 93 can also be formed on a plane not perpendicular to the straight line along which the optical waveguide 94 runs, so-called oblique end face. At this time, the optical functional integrated device 80 exerts an effect of preventing the reflection of the end face, which reduces the light traveling through the optical waveguide 94 from reflecting on the end face of the optical waveguide 94 facing the optical waveguide 93.

While the invention has been described in terms of several embodiments, it is to be understood that these embodiments are merely for the purpose of illustrating the invention and should not be taken as limiting the scope of the claims It will be apparent to those skilled in the art.

This application claims the priority based on Japanese Patent Application No. 2009-004231 filed on January 13, 2009, and incorporates all of the disclosure thereof.

An optical waveguide formed inside the cantilever, an opposite side optical device disposed so as to face the other end of the cantilever, And a heater bonded to the cantilever, wherein the heater is formed of a material having a thermal expansion coefficient different from a thermal expansion coefficient of a material of which the cantilever is formed.

2. The optical waveguide according to claim 1, wherein the heater thermally expands upon application of an electric current, and the cantilever is configured to emit light that travels between the opposed-side optical device and the optical waveguide when the heater thermally expands So that the amount of the optical functional device is changed.

The optical functional integrated device according to claim 2, further comprising a base to be fixed to the opposite side optical device, wherein the base is disposed at a position where the cantilever can contact.

4. The integrated optical device according to claim 3, wherein the amount of light is maximum when the cantilever contacts the pedestal.

The semiconductor device according to claim 2, further comprising a stress control film bonded to the cantilever, wherein the stress control film is formed so that the amount of the light is maximized when the current is not applied to the heater, An optical functional integrated device for stressing said cantilever.

6. The optical functional integrated device according to any one of claims 1 to 5, wherein the opposing optical device is an optical fiber that transmits the light to another device.

6. The optical functional integrated device according to any one of claims 1 to 5, wherein the opposing optical device is a light emitting element that emits the light.

6. The optical functional integrated device according to any one of claims 1 to 5, wherein the opposing optical device is a light receiving element that receives the light.

The optical functional integrated device according to any one of claims 1 to 8, wherein an end of the optical waveguide facing the opposing optical device is formed on an oblique end face.

The optical device according to any one of claims 1 to 9, wherein the opposing optical device comprises another optical waveguide, and an end of the other optical waveguide facing the cantilever is a light formed on an oblique end face Function integrated device.

11. The optical functional integrated device according to any one of claims 1 to 10, further comprising a spot size converter for converting the spot size of the light.

The optical functional integrated device according to any one of claims 1 to 11, further comprising a matching oil filled in a space where the cantilever is disposed.

13. The optical functional integrated device according to any one of claims 1 to 12, further comprising a control circuit for applying the current to the heater so that the amount of light becomes one of the maximum value and the minimum value.

14. The optical functional integrated device according to any one of claims 1 to 13, wherein the heater is joined from the one end of the cantilever to the other end.

A step of fabricating a cantilever in which an optical waveguide is formed and one end of which is supported by a support side portion; a step of fabricating an opposite side optical device disposed so as to face the other end of the cantilever And a step of preparing a heater which is formed of a material having a coefficient of thermal expansion different from a thermal expansion coefficient of a material on which the cantilever is formed and which is bonded to the cantilever beam.

16. The method for manufacturing an optical functional integrated device according to claim 15, further comprising the step of preparing a base to be fixed with respect to the opposite side optical device, wherein the base is disposed at a position where the cantilever can contact.

16. The method according to claim 15, further comprising the step of fabricating a stress control film bonded to the cantilever, wherein the stress control film is formed such that when the current is not applied to the heater, the amount of the light is maximized So as to stress the cantilever beam.

18. The method for manufacturing an optical functional integrated device according to any one of claims 15 to 17, wherein the opposing optical device is an optical fiber that transmits the light to another device.

18. The method for manufacturing an optical functional integrated device according to any one of claims 15 to 17, wherein the opposing optical device is a light emitting element that emits the light.

18. The method for manufacturing an optical functional integrated device according to any one of claims 15 to 17, wherein the opposing optical device is a light receiving element that receives the light.

22. The method for manufacturing an optical functional integrated device according to any one of claims 15 to 20, further comprising forming an end of the optical waveguide facing the opposing optical device on an oblique end face.

22. The method for manufacturing an optical functional integrated device according to any one of claims 15 to 21, further comprising the step of fabricating a spot size converter for converting the spot size of the light.

23. The method for manufacturing an optical functional integrated device according to any one of claims 15 to 22, further comprising a step of filling a space where the cantilever is arranged with a matching oil.

The method for manufacturing an optical functional integrated device according to any one of claims 15 to 23, wherein the heater is bonded from the one end of the cantilever to the other end.